



Task-1

Data Cleaning & Preprocessing

1. Objective

To transform the raw Netflix dataset into a clean, structured format by addressing null values, duplicates, and inconsistent data types using Python (Pandas).

2. Tools Used

- **Language:** Python
- **Library:** Pandas
- **Environment:** Google Colab / Jupyter Notebook

3. Implementation of the Core Concepts

🔗 Identify and Handle Missing Values

Description: Used `.isnull().sum()` to locate missing data. Critical columns like director and cast had significant nulls, which were filled with "Unknown" to maintain data volume.

Coding:

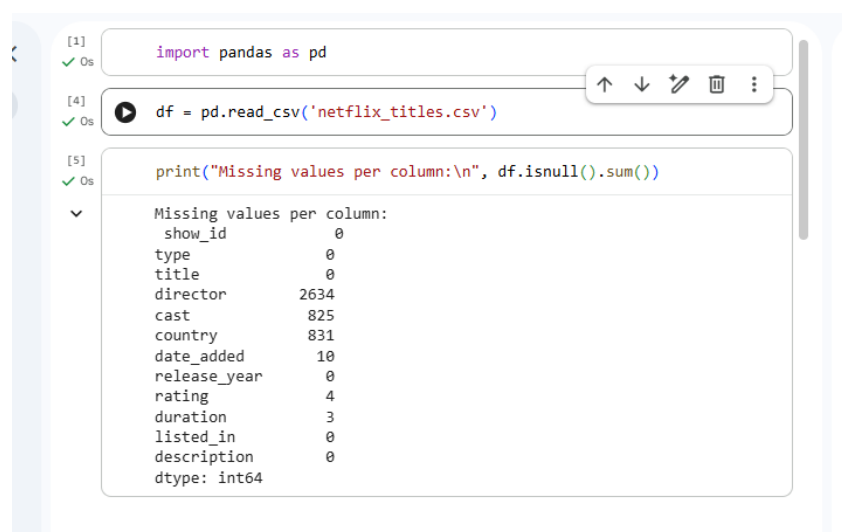
```
import pandas as pd

# Load dataset
df = pd.read_csv('netflix_titles.csv')

# Identify missing values
print("Missing values per column:\n", df.isnull().sum())
```

```
# Fill missing categorical values
df['director'] = df['director'].fillna('Unknown')
df['cast'] = df['cast'].fillna('Unknown')
df['country'] = df['country'].fillna('Not Specified')
# Drop rows where critical info like 'date_added' or 'rating' is missing
df.dropna(subset=['date_added', 'rating', 'duration'], inplace=True)
```

Screenshot:



The screenshot shows a Jupyter Notebook interface with three code cells. The first cell contains `import pandas as pd`. The second cell contains `df = pd.read_csv('netflix_titles.csv')`. The third cell contains `print("Missing values per column:\n", df.isnull().sum())`. The output of the third cell is displayed below the code, showing the number of missing values for each column in the 'netflix_titles.csv' dataset.

Column	Missing values
show_id	0
type	0
title	0
director	2634
cast	825
country	831
date_added	10
release_year	0
rating	4
duration	3
listed_in	0
description	0

dtype: int64

Remove Duplicate Rows

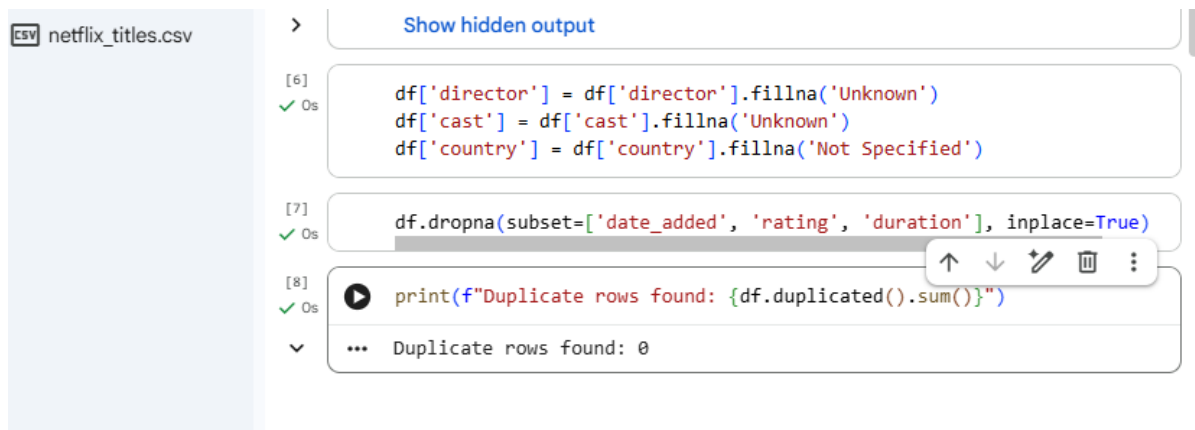
Description: Duplicates can lead to incorrect statistics. I used `.drop_duplicates()` to ensure every row represents a unique title.

Coding:

```
# Check for duplicates
print(f"Duplicate rows found: {df.duplicated().sum()}")

# Remove duplicates
df.drop_duplicates(inplace=True)
```

Screenshot:



```
[6] df['director'] = df['director'].fillna('Unknown')
    df['cast'] = df['cast'].fillna('Unknown')
    df['country'] = df['country'].fillna('Not Specified')

[7] df.dropna(subset=['date_added', 'rating', 'duration'], inplace=True)

[8] print(f"Duplicate rows found: {df.duplicated().sum()}")

... Duplicate rows found: 0
```

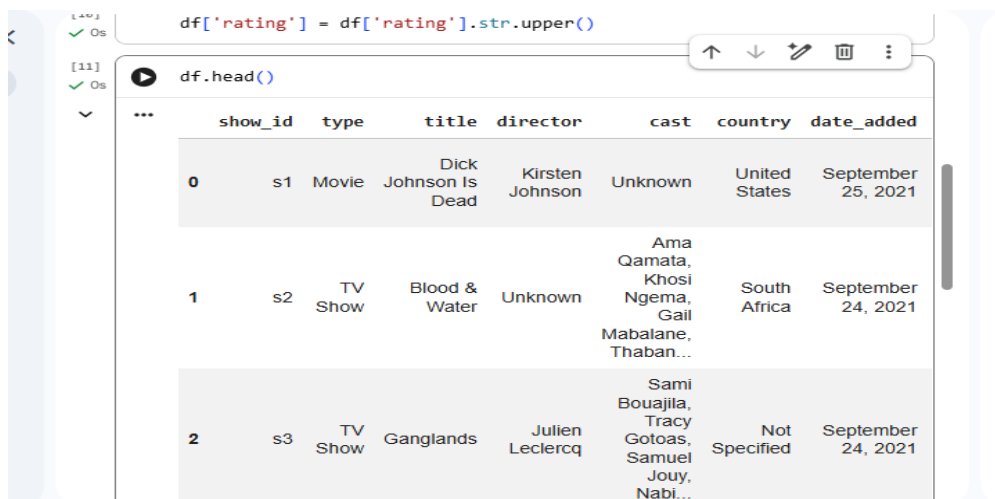
Standardize Text Values

Description: Standardized the type (Movie/TV Show) and rating columns by removing leading/trailing spaces and ensuring consistent casing.

Coding:

```
# Strip extra spaces from the 'type' and 'title' columns
df['type'] = df['type'].str.strip()
df['title'] = df['title'].str.strip()
# Standardize 'rating' to uppercase
df['rating'] = df['rating'].str.upper()
```

Screenshot:



```
[10] df['rating'] = df['rating'].str.upper()

[11] df.head()
```

	show_id	type	title	director	cast	country	date_added
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	September 25, 2021
1	s2	TV Show	Blood & Water	Unknown	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabl...	Not Specified	September 24, 2021

Convert Date Formats

Description: The date_added column was a string. I converted it to a standard datetime format (YYYY-MM-DD) to allow for time-series analysis.

Coding:

```
# Remove leading/trailing spaces from the string date
df['date_added'] = df['date_added'].str.strip()
# Convert to datetime format
df['date_added'] = pd.to_datetime(df['date_added'])
```

Screenshot:



The screenshot shows a Jupyter Notebook interface. The first code cell contains two lines of Python code: `df['date_added'] = df['date_added'].str.strip()` and `df['date_added'] = pd.to_datetime(df['date_added'])`. The second code cell contains `df[['title', 'date_added']].head()`. Below the code, the output is displayed as a table with two columns: 'title' and 'date_added'. The table shows the first five rows of data.

	title	date_added
0	Dick Johnson Is Dead	2021-09-25
1	Blood & Water	2021-09-24
2	Ganglands	2021-09-24
3	Jailbirds New Orleans	2021-09-24
4	Kota Factory	2021-09-24

Rename Column Headers

Description: Renamed headers to follow the snake_case convention (lowercase with underscores) for better compatibility with SQL and coding standards.

Coding:

Rename columns to be clean and uniform

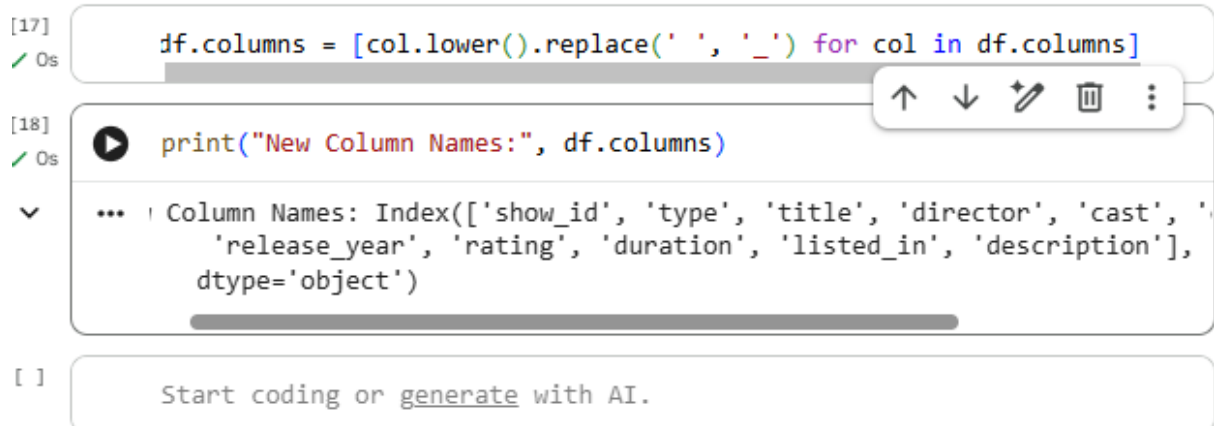
```
df.columns = [col.lower().replace(' ', '_') for col in df.columns]
```

Example: If you want to rename a specific column

```
df.rename(columns={'listed_in': 'category'}, inplace=True)
```

```
print("New Column Names:", df.columns)
```

Screenshot:



The screenshot shows a Jupyter Notebook interface with three cells. The first cell contains the code `df.columns = [col.lower().replace(' ', '_') for col in df.columns]` and has a green checkmark icon. The second cell contains the code `print("New Column Names:", df.columns)` and has a green checkmark icon. The third cell shows the output of the print statement: `Column Names: Index(['show_id', 'type', 'title', 'director', 'cast', 'release_year', 'rating', 'duration', 'listed_in', 'description'], dtype='object')`. Below the cells is a prompt that says "Start coding or generate with AI."

🔗 Check and Fix Data Types

Description: Verified that numeric columns like `release_year` are integers and the newly converted date columns are recognized as datetime objects.

Coding:

Check data types

```
print("Data types before fix:\n", df.dtypes)
```

Ensure `release_year` is an integer

```
df['release_year'] = df['release_year'].astype(int)
```

Final check

```
print("Data types after fix:\n", df.dtypes)
```

Save the cleaned data

```
df.to_csv('cleaned_netflix_data.csv', index=False)
```

```
print("\n--- Project Task 1 Completed Successfully! ---") print(df.head())
```

Screenshots:

```
[19]
✓ Os
print("Data types before fix:\n", df.dtypes)

Data types before fix:
show_id      object
type         object
title        object
director     object
cast         object
country      object
date_added   datetime64[ns]
release_year  int64
rating       object
duration     object
listed_in    object
description  object
dtype: object
```

```
[20]
✓ Os
df['release_year'] = df['release_year'].astype(int)
print("Data types after fix:\n", df.dtypes)

Data types after fix:
show_id      object
type         object
title        object
director     object
cast         object
country      object
date_added   datetime64[ns]
release_year  int64
rating       object
duration     object
listed_in    object
description  object
dtype: object
```

[22]

✓ Os

df.info()

▼

```
... <class 'pandas.core.frame.DataFrame'>
Index: 8790 entries, 0 to 8806
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   show_id                8790 non-null   object
1   type                   8790 non-null   object
2   title                  8790 non-null   object
3   director               8790 non-null   object
4   cast                   8790 non-null   object
5   country                8790 non-null   object
6   date_added             8790 non-null   datetime64[ns]
7   release_year           8790 non-null   int64
8   rating                 8790 non-null   object
9   duration               8790 non-null   object
10  listed_in              8790 non-null   object
11  description             8790 non-null   object
dtypes: datetime64[ns](1), int64(1), object(10)
memory usage: 892.7+ KB
```

[23]

✓ Os

```
print("\n--- Project Task 1 Completed Successfully! ---")
print(df.head())
```

▼

```
...
--- Project Task 1 Completed Successfully! ---
  show_id  type  title  director \
0      s1  Movie  Dick Johnson Is Dead  Kirsten Johnson
1      s2  TV Show  Blood & Water  Unknown
2      s3  TV Show  Ganglands  Julien Leclercq
3      s4  TV Show  Jailbirds New Orleans  Unknown
4      s5  TV Show  Kota Factory  Unknown

                                cast  country \
0                                Unknown  United States
1  Ama Qamata, Khosi Ngema, Gail Mababane, Thaban...  South Africa
2  Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...  Not Specified
3                                Unknown  Not Specified
4  Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...  India

  date_added  release_year  rating  duration \
0  2021-09-25          2020  PG-13    90 min
1  2021-09-24          2021  TV-MA  2 Seasons
2  2021-09-24          2021  TV-MA    1 Season
3  2021-09-24          2021  TV-MA    1 Season
4  2021-09-24          2021  TV-MA  2 Seasons
```

